

Goal: We want to remind ourselves of the structure of $+$, $\cdot$ on $\mathbb{Z}$.

We have 2 operations:

$$\begin{aligned} + : \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} \\ (a, b) &\longmapsto a+b \\ \cdot : \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} \\ (a, b) &\longmapsto a \cdot b \end{aligned}$$

Here are some elementary properties that they satisfy.

SUM:

• Associativity: $a + (b + c) = (a + b) + c$
 $\forall a, b, c \in \mathbb{Z}$

• Existence of additive identity: $\forall a \in \mathbb{Z} \quad a + 0 = 0 + a = a$

• Existence of additive inverses: $\forall a \in \mathbb{Z} \exists b \in \mathbb{Z} \quad a + b = 0$ (take $b = -a$)

• Commutativity: $\forall a, b \in \mathbb{Z} \quad a + b = b + a$.

PRODUCT:

• Associativity: $\forall a, b, c \in \mathbb{Z} \quad a \cdot (b \cdot c) = (a \cdot b) \cdot c$

• Existence of multiplicative identity:
 ~~$\forall a, b \in \mathbb{Z}$~~ $\forall a \in \mathbb{Z} \quad a \cdot 1 = 1 \cdot a = a$.

• Commutativity: $\forall a, b \in \mathbb{Z} \quad a \cdot b = b \cdot a$.

The operations $+$ and $\cdot$ interact by the following law:

• Distributivity: $\forall a, b, c \in \mathbb{R}$

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

For you most of these properties are obvious but they will form the foundation of future definitions: groups and rings.

Remark:^{a)} All of the above holds also for $\mathbb{Q}$. In this case we have an extra property:

$$\left[\begin{array}{l} \text{Given } a \in \mathbb{Q} \setminus \{0\} \exists b \in \mathbb{Q} \text{ such that} \\ ab = ba = 1. \end{array} \right.$$

This extra property will motivate the definition of a field.

b) The significance of the associativity law is that summing and multiplying a finite collection of integers makes sense. i.e. independent of how we do it.

c) Another property that is very special about $\mathbb{Z}$ (and $\mathbb{Q}$) is that the product of two non-zero elements is again non-zero.

More precisely:

For $a, b \in \mathbb{Z}$ such that $ab = 0 \Rightarrow$ either $a = 0$ or $b = 0$.

Later this property will mean that $\mathbb{Z}$ is an integral domain.

↓ consequence

Cancellation law: For $a, b, c \in \mathbb{Z}$

$ca = cb$ and $c \neq 0$ then $a = b$.

FACTORIZATION AND THE FUNDAMENTAL THEOREM OF ARITHMETIC.

Definition: Let $a, b \in \mathbb{Z}$ then a divides b

iff $\exists c \in \mathbb{Z}$ such that $b = ca$. We

denote this by $a|b$ and say that a is a divisor of b . (or factor)

Remarks:

a) 0 is divisible by any integer.

b) 1 is only divisible by 1 and -1

c) Any way of expressing an integer

is called a factorization.

Definition: A prime number p is an integer greater than 1 whose only positive divisors are p and 1.

A positive integer which is not prime is called composite.

Remark: $(\mathbb{Z}, +)$ is generated by 1.

↓

we mean that every integer can be attained by adding successively 1 (or -1) to itself!

- but with multiplication the situation is a bit more complex. There is no single element of $\mathbb{Z}$ that generates $(\mathbb{Z}, \times)$ in the above sense.

a)
Definition: Let $a, b \in \mathbb{Z}$. The highest common factor of a and b , denoted $\text{HCF}(a, b)$ is the largest positive integer which is a common factor of a and b .

b) two non-zero integers are said to be coprime if $\text{HCF}(a, b) = 1$.

A few theorems back to Euclid (300 BC) which I won't prove now since we will later prove them in MORE GENERALITY:

Reminder Theorem: Given $a, b \in \mathbb{Z}$ if $b > 0$ then there is are unique $q, r \in \mathbb{Z}$ s.t. $a = bq + r$ and $0 \leq r < b$.

Theorem: Given $a, b \in \mathbb{Z}$ $\exists u, v \in \mathbb{Z}$ such that $a \cdot u + b \cdot v = \text{H}(a, b)$.

In particular a and b are coprime iff there exist u and v in $\mathbb{Z}$ such that

$$\boxed{au + bv = 1}$$

Euclid's Lemma: Let p be prime and $a, b \in \mathbb{Z}$ then:

$$p \mid ab \Rightarrow p \mid a \text{ or } p \mid b$$

The fundamental theorem of arithmetic:

Every positive integer a greater than 1 can be written as a product of primes.

$$a = p_1 \cdots p_r$$

such factorization is unique up to ordering.

(EXISTENCE)

Proof: We proceed by contradiction

[If there is a positive integer not expressible as a product of primes,]^p

take ~~c~~ $c \in \mathbb{N}$ be the least of such elements.

What do we know about c ?

c cannot be neither 1 or p for p prime.

hence $c = c_1 \cdot c_2$ where $c_1, c_2 \in \mathbb{N}$ and
 $\cdot c_1 < c \quad \cdot c_2 < c$.

By minimality of the choice of c then c_1 and c_2 are products of primes. But then c is expressible as the product of primes! $\nexists$ contradiction.

UNIQUENESS: We want to prove the uniqueness of such decomposition up to ordering.

Assume $a = p_1 \cdots p_r = q_1 \cdots q_m$
 $\quad \quad \quad \swarrow \quad \searrow$
 $\quad \quad \quad$ two decompositions

Then $p_1 \mid q_1 \cdots q_m$ by Euclid's lemma
 $p_1 \mid q_i$ for some i . After renumbering
we may assume $i=1$. Since q_1 is a
prime then $\boxed{p_1 = q_1}$

Applying the cancellation law we
obtain.

$$p_2 \cdots p_r = q_2 \cdots q_m$$

Assume $r < m$. We can continue this process until we have

$$1 = q_{r+1} \cdots q_m$$

This is a contradiction as 1 is not divisible by any prime. Hence $r = m$ and after renumbering $p_i = q_i \forall i$. $\square$

Now we can prove something beautiful.

Theorem: There are infinitely many distinct prime numbers.

Proof: Suppose that there are finitely many distinct primes $p_1, \dots, p_r$.

Consider $C = p_1 \cdot p_2 \cdots p_r + 1$ *

Clearly $C > 1$. By the fundamental theorem of arithmetic C is divisible by at least one prime say p_1 .

Then $C = p_1 \cdot d$ for some $d \in \mathbb{Z}$

Then $c = p_1 \cdot d$ for some $d \in \mathbb{Z}$. Hence we have

$$p_1 (d - p_2 \dots p_r) = c - p_1 p_2 \dots p_r = 1 \quad \text{by *}$$

This is a contradiction as no prime divides 1.

Hence there are infinitely many primes.

Q: We have seen that both $\mathbb{Z}$ and $\mathbb{Q}$ are examples of sets with two concepts of composition (+ and $\times$) which satisfy abstract conditions. And we have seen that $(\mathbb{Z}, \times)$ is very rich.

Can we think of other examples of sets with a concept of + and $\circ$ which satisfy the same elementary properties?

Congruences:

Fix $m \in \mathbb{N}$. By the remainder theorem if $a \in \mathbb{Z}$ $\exists!$ $q, r \in \mathbb{Z}$ such that

$$\boxed{a = q \cdot m + r} \quad \text{where} \quad 0 \leq r < m.$$

We call r The remainder of a modulo m .

This gives a natural equivalence relation on $\mathbb{Z}$
 $a \sim b$ iff a and b have the same remainder
modulo m

$$\text{iff } m \mid (a-b)$$

^{a)}
Remarks: The equivalence classes of $\mathbb{Z}$
under this relation are the remainders
modulo m .

Hence there are m distinct equivalence
classes which we call residue classes

We denote the set of residue
classes as

$$\mathbb{Z}/m\mathbb{Z} = \{[0], [1], \dots, [m-1]\}.$$

b) there is a natural surjection map:

$$[\]: \mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z}$$

$$a \longrightarrow [a]$$

Note that this is clearly not injective
as many integers have the same
remainder modulo m .

And we can even put structure on these sets:

Proposition: Let $m \in \mathbb{N}$. Then for all $a, b, a', b' \in \mathbb{Z}$:

$$\begin{array}{ll} \text{if } [a] = [a'] & \text{then } [a+b] = [a'+b'] \\ \text{and } [b] = [b'] & \text{and } [a \cdot b] = [a' \cdot b']. \end{array}$$

Left as an ex.

↓

It allows us to define addition and multiplication on $\mathbb{Z}/m\mathbb{Z}$ by:

product $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z}$

• $([a], [b]) \longmapsto [a \cdot b]$

sum $([a], [b]) \longmapsto [a+b]$

since the structure on $\mathbb{Z}/m\mathbb{Z}$ of $+$ and $\cdot$ is inherited from $\mathbb{Z}$, then $\cdot$ and $+$ also satisfy the elementary properties we already have for $+$ and $\cdot$ in $\mathbb{Z}$

In particular:

a) $[0] \in \mathbb{Z}/m\mathbb{Z}$ behaves like $0 \in \mathbb{Z}$.

$$[0] + [a] = [a] + [0] = [a] \quad \forall [a] \in \mathbb{Z}/m\mathbb{Z}$$

b) $[1] \in \mathbb{Z}/m\mathbb{Z}$ behaves like $1 \in \mathbb{Z}$.

$$[1] \cdot [a] = [a] \cdot [1] = [a] \quad \forall [a] \in \mathbb{Z}/m\mathbb{Z}$$

c) We say that $[a] \in \mathbb{Z}/m\mathbb{Z}$ is non zero if $[a] \neq [0]$. Even though $+$ and $\cdot$ on $\mathbb{Z}/m\mathbb{Z}$ share the same elementary properties as $+$ and $\cdot$ in $\mathbb{Z}$.

They behave quite different.

Examples:

$$\begin{aligned} & \bullet [1] + [1] + \dots + [1] \text{ (m-times)} \\ & = [m] = [0] \end{aligned}$$

• If m is a composite $m = r \cdot s$ for $1 < r, s < m$ then $[r]$ and $[s]$ are non-zero in $\mathbb{Z}/m\mathbb{Z}$ but $[r] \times [s] = [m] = [0] \in \mathbb{Z}/m\mathbb{Z}$.

thus we can have 2 non-zero elements whose product is 0.

Proposition: For every $m \in \mathbb{N}$, $a \in \mathbb{Z}$ the congruence $ax \equiv 1 \pmod{m}$ has a solution (in $\mathbb{Z}$) iff a and m are coprime.

Proof: This is just a restatement of the fact that if a and m are coprime $\exists u, v \in \mathbb{Z}$ s.t. $au + mv = 1$.

Corollary: All non-zero elements of $\mathbb{Z}/m\mathbb{Z}$ have a multiplicative inverse $\Leftrightarrow m$ is prime.

later: this will be re-stated as $\mathbb{Z}/m\mathbb{Z}$ is a field iff m is prime.

KEY OBS: there are natural sets (like $\mathbb{Z}$ and $\mathbb{Q}$) which come equipped with a concept of $+$ and $\cdot$ whose most basic properties are the same as those of the usual addition and multiplication.

Don't be fooled into thinking all

other examples will come from numbers.
As we'll see there are many more
exotic examples.

Appendix: let $m \in \mathbb{N}$.

and consider the relation: on $\mathbb{Z}$

$$x \sim y \iff m \mid (x - y)$$

$\sim$ is called the congruence modulo m .

show that $\sim$ is an eq relation on $\mathbb{Z}$.

reflexivity: let $z \in \mathbb{Z}$ we want to show

$$z \sim z \text{ i.e. } m \mid (z - z) \quad z - z = 0$$

and 0 is divisible by m . so $(z, z) \in E_m$

symmetry: Assume $z, w \in \mathbb{Z}$ $(z, w) \in E_m$

we want to show $(w, z) \in E_m$.

$$\text{if } (z, w) \in E_m \iff m \mid (z - w)$$

$$\iff \exists b \in \mathbb{Z} \text{ s.t. } z - w = m \cdot b$$

multiply both sides by -1

$$\text{then } w - z = m \cdot (-b)$$

thus $w - z$ is divisible by m .

Then $(w, z) \in E_m$.

transitivity: Suppose $(z, w) \in E_m$ and $(w, l) \in E_m$.

We want to show $(z, l) \in E_m$ i.e.
 $m \mid (z - l)$.

since $(z, w) \in E_m \exists b \in \mathbb{Z} \ z - w = m \cdot b$

since $(w, l) \in E_m \exists a \in \mathbb{Z} \ w - l = m \cdot a$.

$$\begin{aligned} \text{then } \underbrace{(z - w) + (w - l)}_{z - l} &= \underbrace{mb + ma}_{= m(b+a)} \\ &\quad \underbrace{}_{\in \mathbb{Z}} \end{aligned}$$

thus $(z, l) \in E_m$.

Lecture: ~~Groups and Examples~~

Binary operations and examples

Solaris: We are familiar with addition and multiplication in the real numbers

↳ we have thought of them as operations with a notion of composition.

We consider $+$ and $\cdot$ to be binary operations

Our goal for the course abstract this notion and study sets in which we have one or more operations, and study the structural properties.



what do I mean?

different:

- in $(\mathbb{R}, +)$ the equation $x+x=a$ for $a \in \mathbb{R}$ has a solution, take $x = \frac{a}{2}$.
- in $(\mathbb{R}, \cdot)$ the same equation $x \cdot x = a$ does not always have a solution ($a = -1$).

Sometimes two different sets with operations that we naturally regard as different turn out to be structurally the same!

↳ idea of isomorphism: e.g. $(\mathbb{R}, +)$
 $(\mathbb{R}^2, +)$

Idea: We want to have a distinguished set S equipped with a binary operation $\star: S \times S \rightarrow S$, and study its structure.

Definition: (Binary operation) Let S be a set a binary operation on S is a function mapping $\star: S \times S \rightarrow S$.
 $(a, b) \mapsto a \star b$.

Examples:

$\bullet (\mathbb{Z}, +)$	$\bullet (\mathbb{R}, +)$
$\bullet (\mathbb{Z}, \cdot)$	$\bullet (\mathbb{R}^n, \cdot)$
$\bullet (\mathbb{Q}^*, \cdot)$	$\bullet (\mathbb{N}, +)$
$\bullet (\mathbb{Q}, +)$	$\bullet (\mathbb{N}, \cdot)$

More exotic examples:

1) the set of Matrices of $n \times n$ with addition and entries in the real numbers.

$M_{n \times n}(\mathbb{R})$ equipped with the sum:

$$M_{n \times n}(\mathbb{R}) \times M_{n \times n}(\mathbb{R}) \longrightarrow M_{n \times n}(\mathbb{R})$$

$$(A, B) \longrightarrow A+B.$$

~~It is~~ Then $(M_{n \times n}(\mathbb{R}), +)$ is a set with a binary operation.

If instead of $M_{n \times n}(\mathbb{R})$ we take all matrices with entries on $\mathbb{R}$ then the sum is not longer ~~an~~ a binary operation.

↳ Indeed $\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$ I have no idea how to compute.
 $\uparrow$
 fails to be a function.

~~How else could be a w~~

b) $GL_n(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid A \text{ is invertible}\}.$

Then $GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \longrightarrow GL_n(\mathbb{R})$

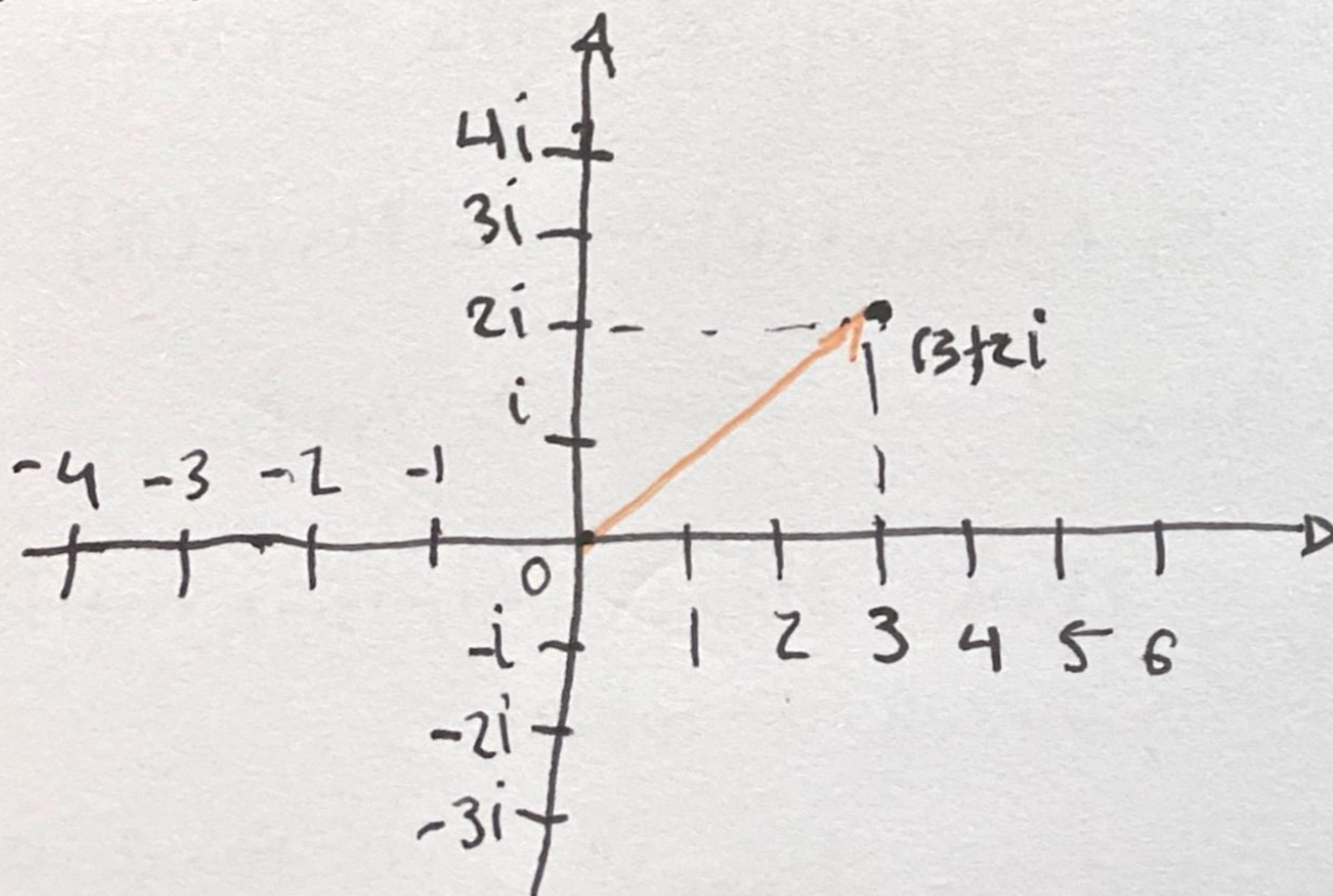
$(A, B) \longrightarrow A \cdot B$

is a binary operation.

c) the complex numbers: $\begin{cases} \rightarrow a \text{ is the real part.} \\ \rightarrow b \text{ is the imaginary part.} \end{cases}$

Recall that $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$

and we imagine them as a point in an euclidean plane.



they arise to give a solution to a quadratic equation $x^2 = -1$, and i is the solution i.e. $i^2 = -1$.

we call i an imaginary element.

so we can consider $\mathbb{C}$ with the binary operation given by multiplication.

$$(a+ib)(c+id) = (ac-bd) + (ad+bc)i.$$

why? just compute.

$$\begin{aligned}(a+ib)(c+id) &= ac + iad + ibc + i^2 bd \\ &= ac + iad + ibc - bd. \\ &= (ac-bd) + i(ad+bc).\end{aligned}$$

Definition: Absolute value of $a+ib$ is

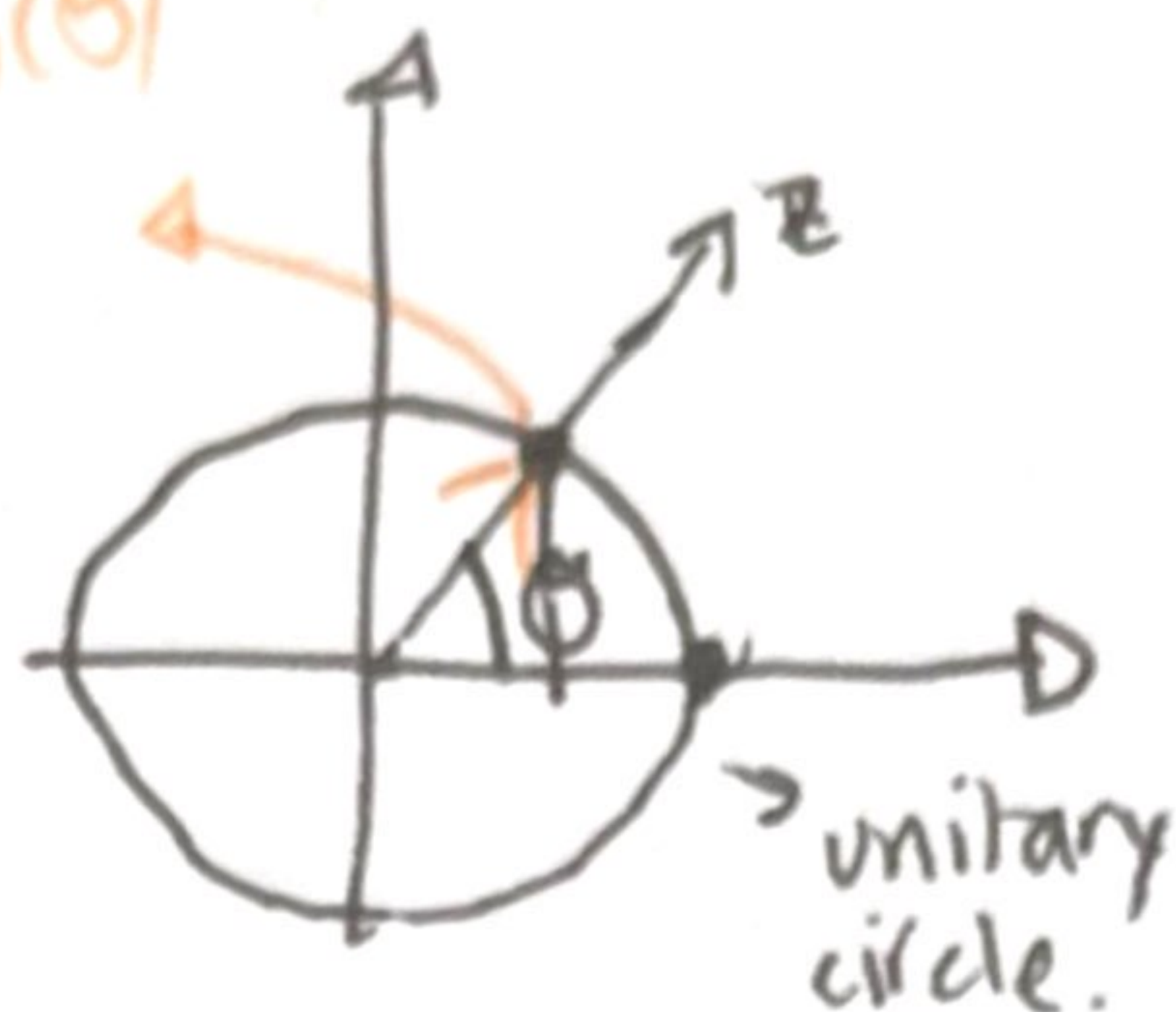
$$|a+ib| = \sqrt{a^2 + b^2}$$

the length of the vector.

We can think of complex numbers in a different way (in POLAR COORDINATE FORM) $z = |z| \cdot (\cos(\theta) + i \sin(\theta))$

and scale by the norm,

$$\cos\theta + i\sin(\theta)$$



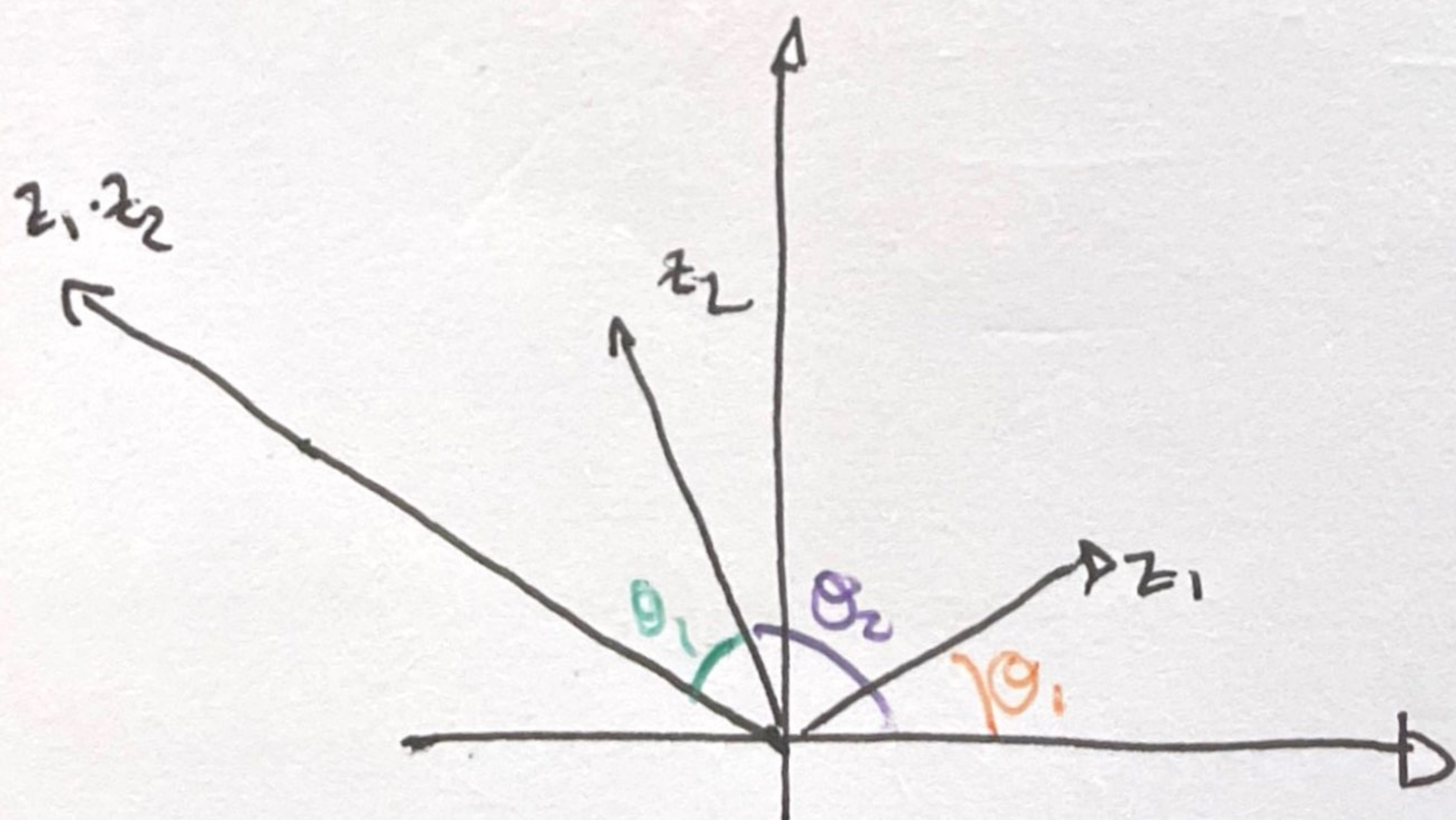
Euler's formula $e^{i\theta} = \cos\theta + i\sin(\theta)$

Allow us to think about multiplication in $\mathbb{C}$ another way.

$$z_1 \cdot z_2 = (|z_1| e^{i\theta_1}) \cdot (|z_2| e^{i\theta_2}) = |z_1| |z_2| e^{i(\theta_1 + \theta_2)}$$

$$z_1 = |z_1| e^{i\theta_1}$$

$$z_2 = |z_2| e^{i\theta_2}$$



$$(\mathbb{C}, \cdot)$$

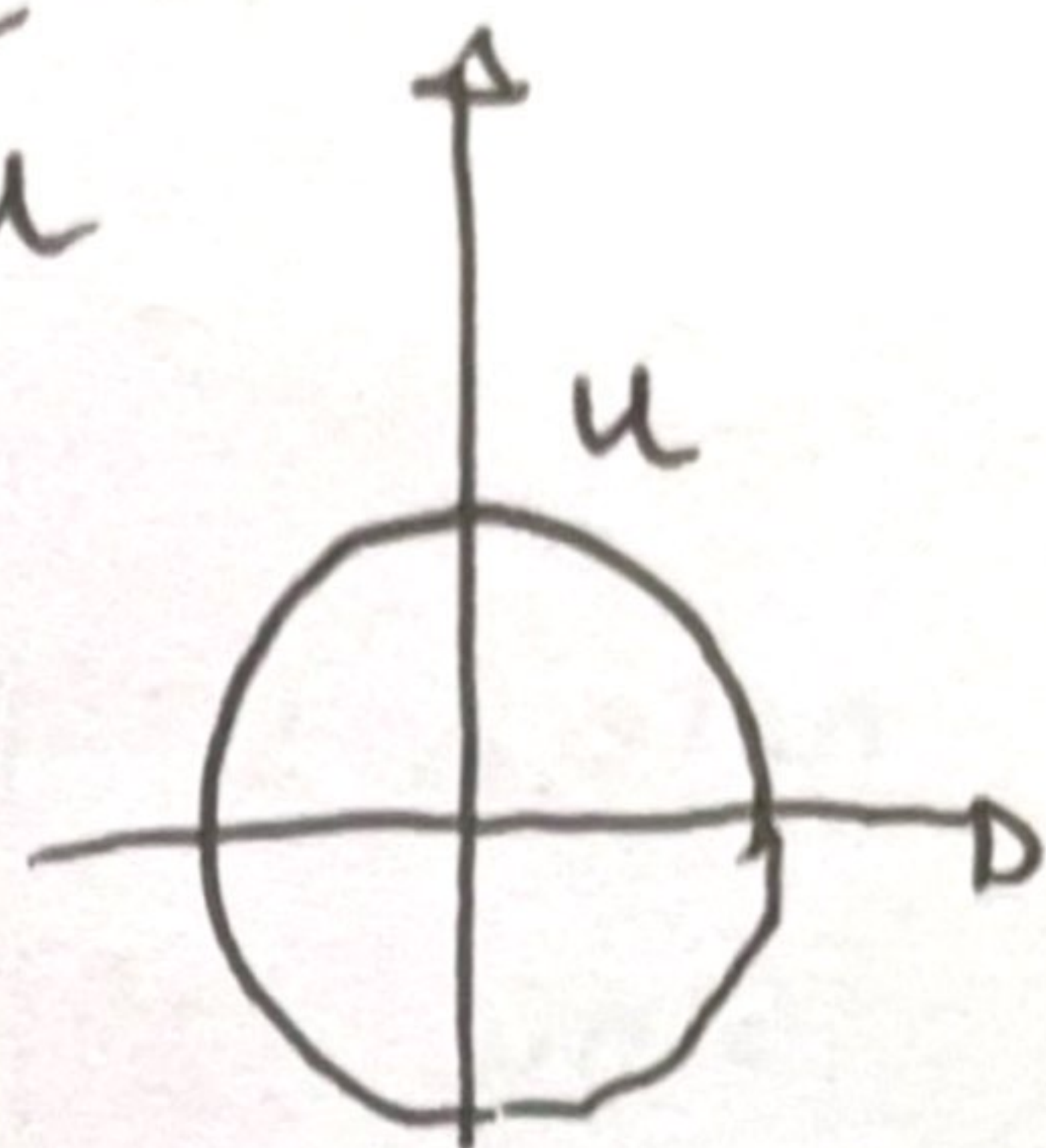
d) the ^{unit circle} ~~roots of unity~~ with multiplication.

$$\text{Let } U = \{z \in \mathbb{C} \mid |z| = 1\}$$

then note that I can take
also the multiplication on U
and it still makes sense.

$$U \times U \longrightarrow U$$

$$(z_1, z_2) \longrightarrow z_1 \cdot z_2$$



$$z_1 = |z_1| e^{i\theta_1} \quad \text{then } z_1 \cdot z_2 = |z_1| |z_2| e^{i(\theta_1 + \theta_2)}$$

$$z_2 = |z_2| e^{i\theta_2}$$

but since we are restricted to U :

$$z_1 = e^{i\theta_1} \text{ since } |z_1| = 1$$

$$z_2 = e^{i\theta_2} \text{ since } |z_2| = 1$$

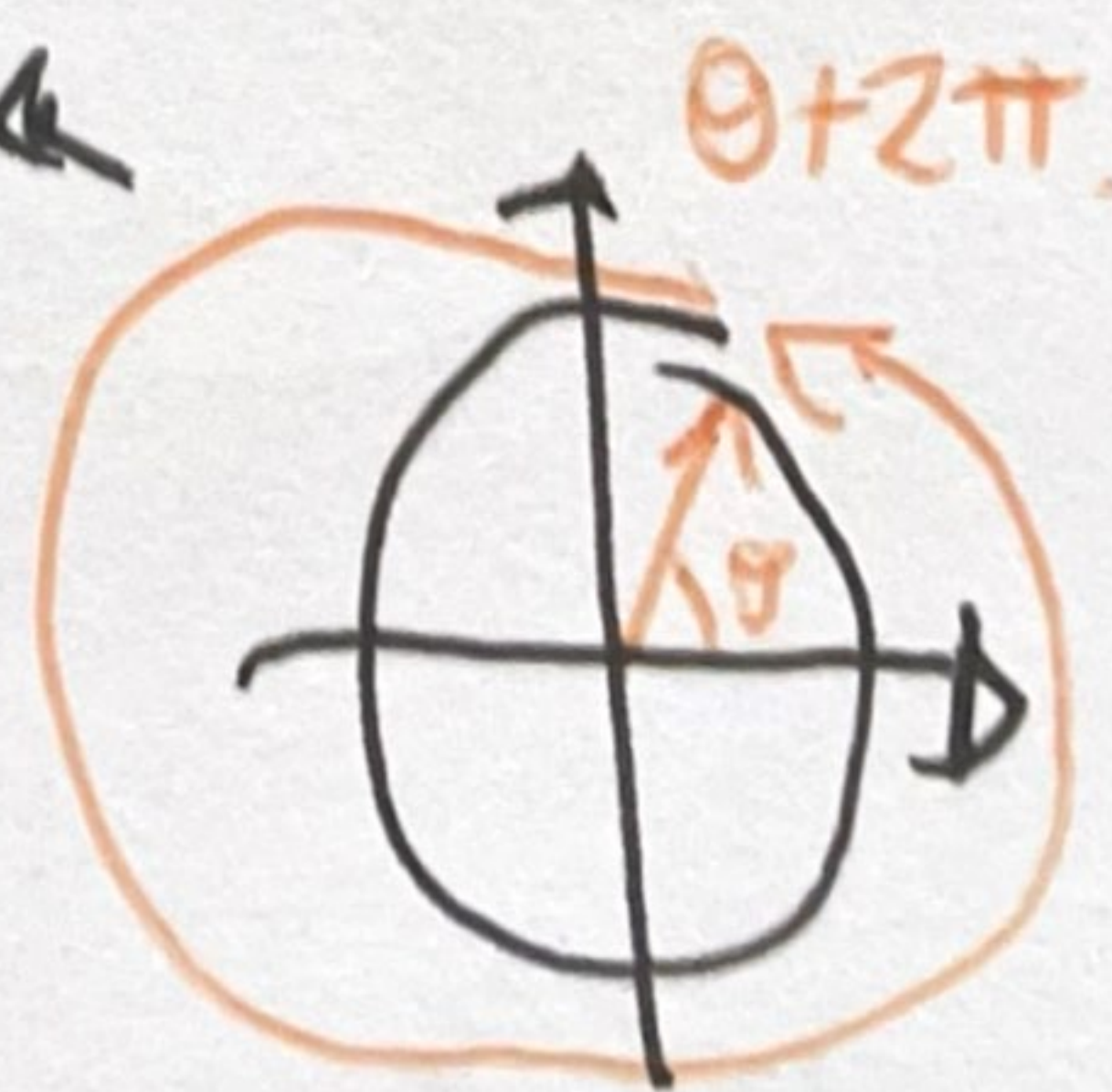
$$\text{so } z_1 \cdot z_2 = e^{i(\theta_1 + \theta_2)} \rightarrow \text{again of norm 1.}$$

warning: we want to be slightly careful

we want to identify them $\leftarrow$

so we associate

$$z_1 \cdot z_2 = e^{i(\theta_1 + \theta_2 \bmod 2\pi)}$$



$$\text{thus } U \times U \longrightarrow U$$

$$(e^{i\theta_1}, e^{i\theta_2}) \longrightarrow e^{i(\theta_1 + \theta_2 \bmod 2\pi)}$$

is a set
with an
operation.

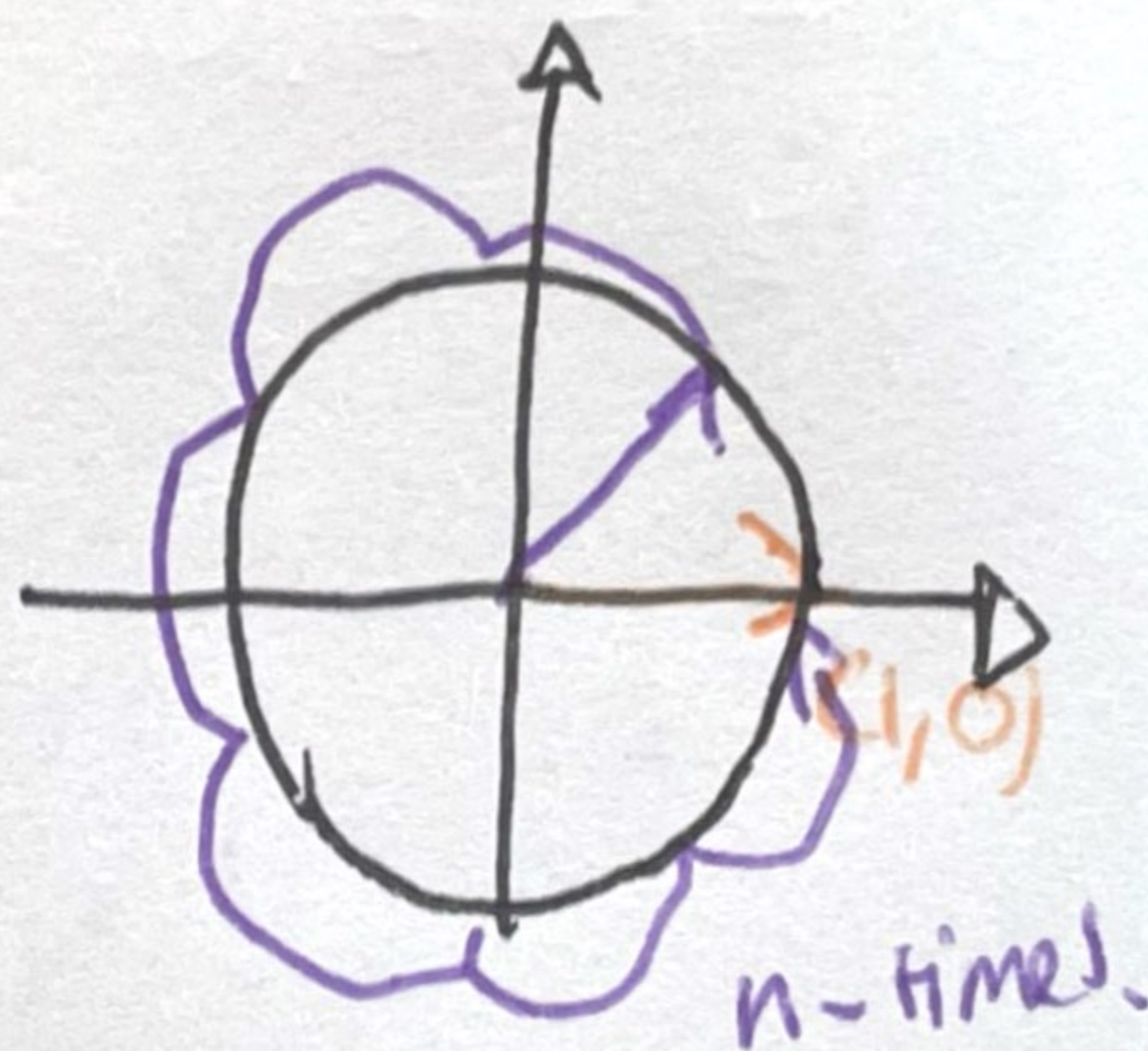
e) Roots of unity are closed under multiplication.

Let $U_n = \{z \in \mathbb{C} \mid z^n = 1\}$ ← are called the n^{th} roots of unity.

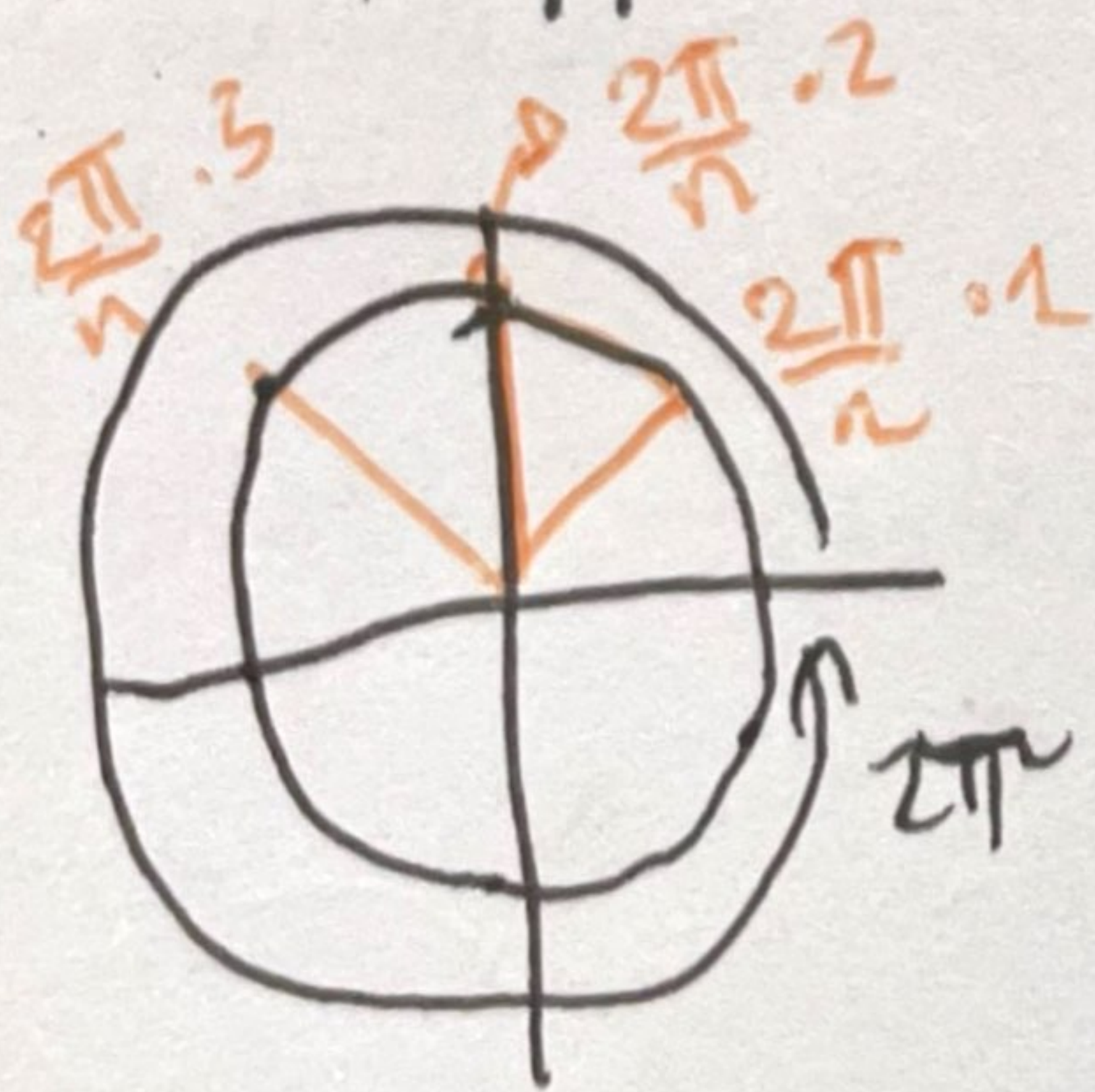
Let's first try to describe them.

note first that if $z \in U_n$ then $|z|^n = 1$

since $|z^n| = 1 \Rightarrow |z|^n = 1 \Rightarrow |z| = 1$.



An intuitive approach:



split it in n -equal parts

$$\frac{2\pi}{n}$$

$$\frac{2\pi}{n} \cdot l \quad \text{for } i \quad 0 \leq l \leq n-1$$

but then

$$\left(e^{i \left(\frac{2\pi}{n} l \right)} \right)^n = e^{i \cdot 2\pi l}$$

$$= \cos(2\pi l) + i \sin(2\pi l)$$

so the roots of unity are ~~not~~ of them.

* all of the form:

$$\cos\left(\ell \cdot \frac{2\pi}{n}\right) + i \sin\left(\ell \cdot \frac{2\pi}{n}\right)$$

for $\ell = 0, \dots, n-1$.

And I can consider all the n^{th} -roots of unity $1 = \zeta^0, \zeta^1, \zeta^2, \dots, \zeta^{n-1}$

since $\zeta^n = 1$ they are closed under multiplication

e.g. $n=10$ $\zeta^6 \cdot \zeta^8 = \zeta^{14} = \zeta^{10} \cdot \zeta^4 = 1 \cdot \zeta^4$

in general $\zeta^i \cdot \zeta^j = \zeta^{i+j \bmod n}$.

Definition: Let $*$ be a binary operation on S and let H be a subset of S . The subset H is closed under ~~multiplication~~ $*$ iff for all $a, b \in H$ we also have $a * b \in H$.

In this case, the binary operation on H given by the restriction of $*$ to H is the induced operation of $*$ on H .

Examples: • let $*$ = + $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$
 $+ (a, b) \rightarrow a + b$

• $H = \mathbb{Z}$ Then $*$ = + $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$
 $(a, b) \rightarrow a + b$

then H is closed.

NON EXAMPLE:

while: take $S = \mathbb{R}$

$$- : \mathbb{R} \rightarrow \mathbb{R}$$

$$x \rightarrow -x$$

Then $H = \mathbb{R}^{>0}$ is NOT closed!

More examples:

a) take $\Phi = S$ with the multiplication.

Then $H = \{1\}$

$$\bullet H = U_n = \{z \in \mathbb{C} \mid |z| = 1 \text{ and } z^n = 1\}$$

$$\bullet H = U = \{z \in \mathbb{C} \mid |z| = 1\}$$

are both closed under multiplication.

b) Let F be the set of real valued functions having as domain the set $\mathbb{R}$ of real numbers.

consider the operations:

$$+ : F \times F \rightarrow F$$

$$(f, g) \rightarrow f+g : x \rightarrow f+g(x)$$

$$\bullet (f, g) \rightarrow f \cdot g : x \rightarrow f(x)g(x)$$

$$- : F \rightarrow F$$

$$f \rightarrow -f : x \rightarrow -f(x)$$

and composition: $F \times F \rightarrow F$
 $(f, g) \rightarrow f \circ g$

All of these functions are again real valued functions with domain $\mathbb{R}$, then F is closed under $+$, $\cdot$, $-$, o .

c) on $\mathbb{Z}^+$ we define a binary operation $*$ by $a * b = \begin{cases} \min\{a, b\} & \text{if } a \neq b \\ a & \text{if } a = b. \end{cases}$

Properties of binary operations.

Def 1: A binary operation $*$ on a set S is commutative if and only if

$$a * b = b * a \quad \text{for every } a, b \in S$$

Examples

$$(\mathbb{Q}, \cdot)$$

$$(\mathbb{U}, \cdot)$$

$$(\mathbb{U}_n, \cdot)$$

$$(\mathbb{F}, *) \quad * = +, \cdot$$

Non examples

$$(\mathbb{F}, o)$$

$$(M_{n \times n}(\mathbb{R}), \cdot)$$

Def 2: $(S, *)$ ($*$ is associative) A binary operation on a set S is associative if $\forall a, b, c \in S$

$$(a * b) * c = a * (b * c)$$

Examples:

• $(\mathbb{R}, +)$ $(a + b) + c = a + (b + c)$

• $(\mathbb{Z}, \cdot)$

Proof: We want to show that given real valued functions f, g and h we have

$$(f \circ g) \circ h = f \circ (g \circ h)$$

To show that 2 functions are equal it is sufficient to show that ~~if $x \in \mathbb{R}$~~ for every $x \in \mathbb{R}$ their values at x are the same.

$$(f \circ g) \circ h(x) = (f \circ g)(h(x)) = f(g(h(x)))$$

and

$$f \circ (g \circ h)(x) = f(g \circ h(x)) = f(g(h(x)))$$

Recall

$$f \circ g(x) = f(g(x))$$

Lecture 6: Isomorphic binary structures

Leading Question: Let say we have 2 sets with binary operations $(S, *)$ and $(S', *)'$ when are them the same structure? How can we determine that they are structurally the same or not? or different?

A motivating example:

let $S = \{a, b, c\}$ with $*$ as

$*$	a	b	c
a	c	a	b
b	a	b	c
c	b	c	a

And $X = \{x, y, z\}$ with $*$ as

$*$	x	y	z
x	z	x	y
y	x	y	z
z	y	z	x

we'll send $\begin{matrix} a \rightarrow x \\ b \rightarrow y \\ c \rightarrow z \end{matrix}$ we are just renaming thing.

$$a * b = a$$



$$f(a) * f(b) = f(a)$$

$$x * y = x$$

so everything we did is just renaming things

Definition: Let $(S, *)$ and $(S', \star')$ be sets with a binary operation we say that there are ISOMORPHIC if there is $f: S \rightarrow S'$ s.t:

a) f is a bijection,

b) For every $x, y \in S$

$$f(x * y) = f(x) *' f(y)$$

HOW TO SHOW THAT TWO BINARY STRUCTURES ARE ISOMORPHIC?

Step 1: Define the function f that gives the isomorphism of S and S' $\rightarrow$ this means we need to describe what $f(s)$ is for every $s \in S$.

Step 2: Show that f is one to one;

this is suppose that $f(x) = f(y)$ in S' and conclude that $x = y$ in S .

Step 3: Show that f is onto S' ; that is suppose that $s' \in S'$ you need to find an $s \in S$ s.t. $f(s) = s'$.

step 4: show that $f(x * y) = f(x) *' f(y)$ for all $x, y \in S$.

This is just a matter of computation, compute both sides of the equation and see whether they are the same.

Example 1: $(\mathbb{R}, +)$ and $(\mathbb{R}^+, \cdot)$

S1 $\left[\begin{array}{l} \text{let } f: (\mathbb{R}, +) \longrightarrow (\mathbb{R}^+, \cdot) \\ r \longrightarrow e^r. \end{array} \right.$

S2 $\left[\begin{array}{l} \text{injective: assume } e^{r_1} = e^{r_2} \text{ then} \\ \log e^{r_1} = \log e^{r_2} \Rightarrow r_1 = r_2 \end{array} \right.$

S3 $\left[\begin{array}{l} \text{onto: take } z \in \mathbb{R}^+ \text{ we want to find } r \text{ s.t.} \\ z = e^r \end{array} \right.$

take $r = \log(z)$ then
$$e^r = e^{\log(z)} = z$$

comp.
$$\begin{array}{l} z = e^r \\ \log(z) = r. \end{array}$$

S4: let $x, y \in (\mathbb{R}, +)$ we want to show

$$f(x+y) = f(x) \cdot f(y)$$

$$f(x+y) = e^{x+y}$$

$$f(x) \cdot f(y) = e^x \cdot e^y$$

And $e^{x+y} = e^x \cdot e^y$ by properties of exponents.

Instead we want to consider **STRUCTURAL PROPERTIES** $\rightarrow$ it is a property P s.t. that if it holds in $(S, *)$ then it holds on every structure isomorphic to $(S, *)$

thus to show: that a property P is a structural property you need to:

STEP 1: Assume $(S, *)$ has it and take $(S', *')$ and arbitrary structure isomorphic to $(S, *)$.

STEP 2: Prove that $(S', *')$ is also has the property.

Examples:

1. the set has 4 elements. $\rightarrow f$ is a bij.

~~2. the operation is comm~~

2. the operation is commutative:

Assume $(S, *)$ $*$ is comm. And let $f: S \rightarrow S'$ be an isomorphism. We aim to show that $*$ ' is commutative in S' .

take $z_1, z_2 \in S'$ goal: $z_1 *' z_2 = z_2 *' z_1$

Since f is onto they are $w_1, w_2 \in S$ s.t. $f(w_1) = z_1$

then since $*$ is commutative in S $f(w_2) = z_2$

we have $w_1 * w_2 = w_2 * w_1$ applying f

in both sides $f(w_1 * w_2) = f(w_2 * w_1)$
 by hom $f(w_1) *' f(w_2) = f(w_2) *' f(w_1)$
 $z_1 *' z_2 = z_2 *' z_1$

then this shows that $(M_{2 \times 2}(\mathbb{R}), \circ) \neq (\mathbb{R}^*, \cdot)$

Example 3: The equation $a * x = b$ has a solution for any $a, b \in S$.

Proof: S1: Assume For every $a, b \in S$ there is x s.t. $a * x = b$. And let $f: (S, *) \rightarrow (S', *')$ be an isomorphism.

S2: Goal: We want to show that $\forall z_1, z_2 \in S'$ there is $y \in S'$ s.t. $z_1 *' y = z_2$.

By surjectivity they are $a, b \in S$ s.t. $f(a) = z_1$ and $f(b) = z_2$.

let x be the sd in S so $a * x = b$.

Apply f to this eq $f(a * x) = f(b)$

so $f(x)$ is the y we are looking for.

$$\begin{array}{ccc} \text{hom} & \parallel & \parallel \\ f(a) *' f(x) & = & z_2 \\ \parallel & & \\ z_1 *' f(x) & = & z_2 \end{array}$$

~~then $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$ are not isomorphic~~

Example 4: the eq $x * x = a$ has a sol.

SA: Assume $\forall a \in S$ $x * x = a$ has a sol.

And let $f: (S, *) \rightarrow (S', *')$ be an isomorphism

SL: take $b \in S'$ we want to show that there is
 $y \in S'$ s.t. $y *' y = b$.

By surj. $\exists a \in S$ st $f(a) = b$.

Since P holds in S $\exists w$ s.t. $w^2 = a$.

i.e. $w * w = a$.

Apply f $\underbrace{f(w * w)} = f(a) = b$.

now $f(w) *' f(w) = b$

thus $y *' y = b$ has a sol.

a) Then $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$ are not isomorphic

in $(\mathbb{C}, +)$ there is x s.t. $x^2 = 1$
while not in $(\mathbb{R}, +)$

b) $(\mathbb{Q}, +)$ and $(\mathbb{Z}, +)$ are NOT isomorphic.

Definition: Let $(S, *)$ be a binary operation on a set we say that it has an identity element if $\exists e \in S$ s.t. $\forall s \in S$
 $s * e = e * s = s$.

$$\underline{e=0} \text{ in } (\mathbb{Z}, +) \quad (\mathbb{R}, +) \quad (\mathbb{Q}, +)$$

$$\underline{e=1} \text{ in } (\mathbb{Q}^{\times}, \cdot)$$

$$e=1 \text{ in } (U_n, \cdot)$$

$$e=I. \quad (M_{n \times n}(\mathbb{R}), \cdot)$$

Theorem: A binary structure $(S, *)$ has at most one identity element.

take e and e' we want to show $e = e'$
identity elements.

$$\begin{array}{c} = e * e' = e' * e = e' \\ \uparrow \qquad \qquad \qquad \uparrow \\ \text{because } e' \text{ is an identity} \quad \text{because } e \text{ is an identity.} \end{array}$$

Theorem 2: If e is an identity element in $(S, *)$ and $f: (S, *) \rightarrow (S', \star')$ is an isomorphism then $f(e)$ is an identity element of $(S', \star')$.

take $s' \in S$ by onto $\exists s \in S$ s.t. $f(s) = s'$.

We have $s * e = e * s = s$ for every $s \in S$

then apply f $f(s * e) = f(e * s) = f(s)$

hom $\downarrow$

hom $\downarrow$ "

"

$$f(s) * f(e) = f(e) * f(s) = s'$$

"

$$s' * f(e) = f(e) * s' = s'$$

as required.